NVIDIA企业级产品及HPC/AI应用

NVIDIA

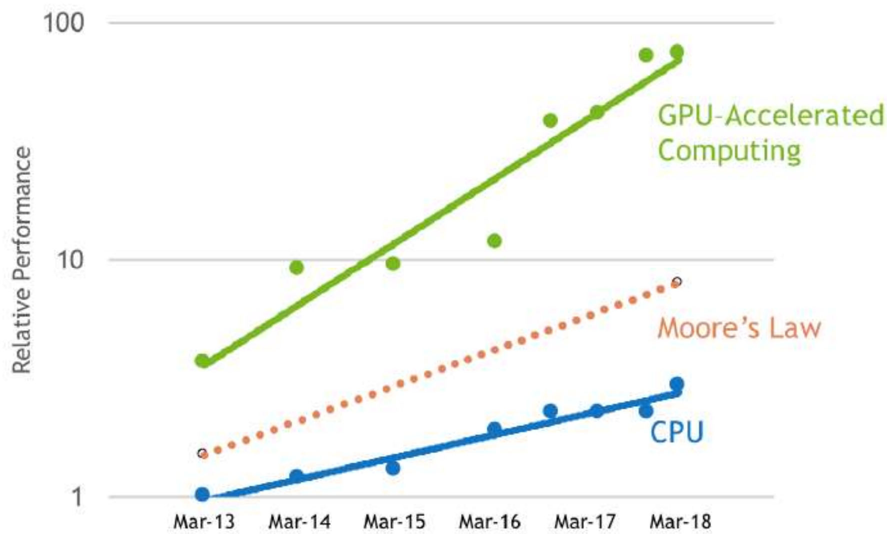
Stephen Wu

超越摩尔定律

2013

- cuBLAS: 5.0
- cuFFT: 5.0
- cuRAND: 5.0
- cuSPARSE: 5.0
- NPP: 5.0
- Thrust: 1.5.3
- CUDA: 5.0
- Resource Mgr: r304
- Base OS: CentOS 6.2

Accelerated Server
With Fermi



Measured performance of Amber, CHROMA, GTC, LAMMPS, MILC,
NAMD, Quantum Espresso, SPECFEM3D

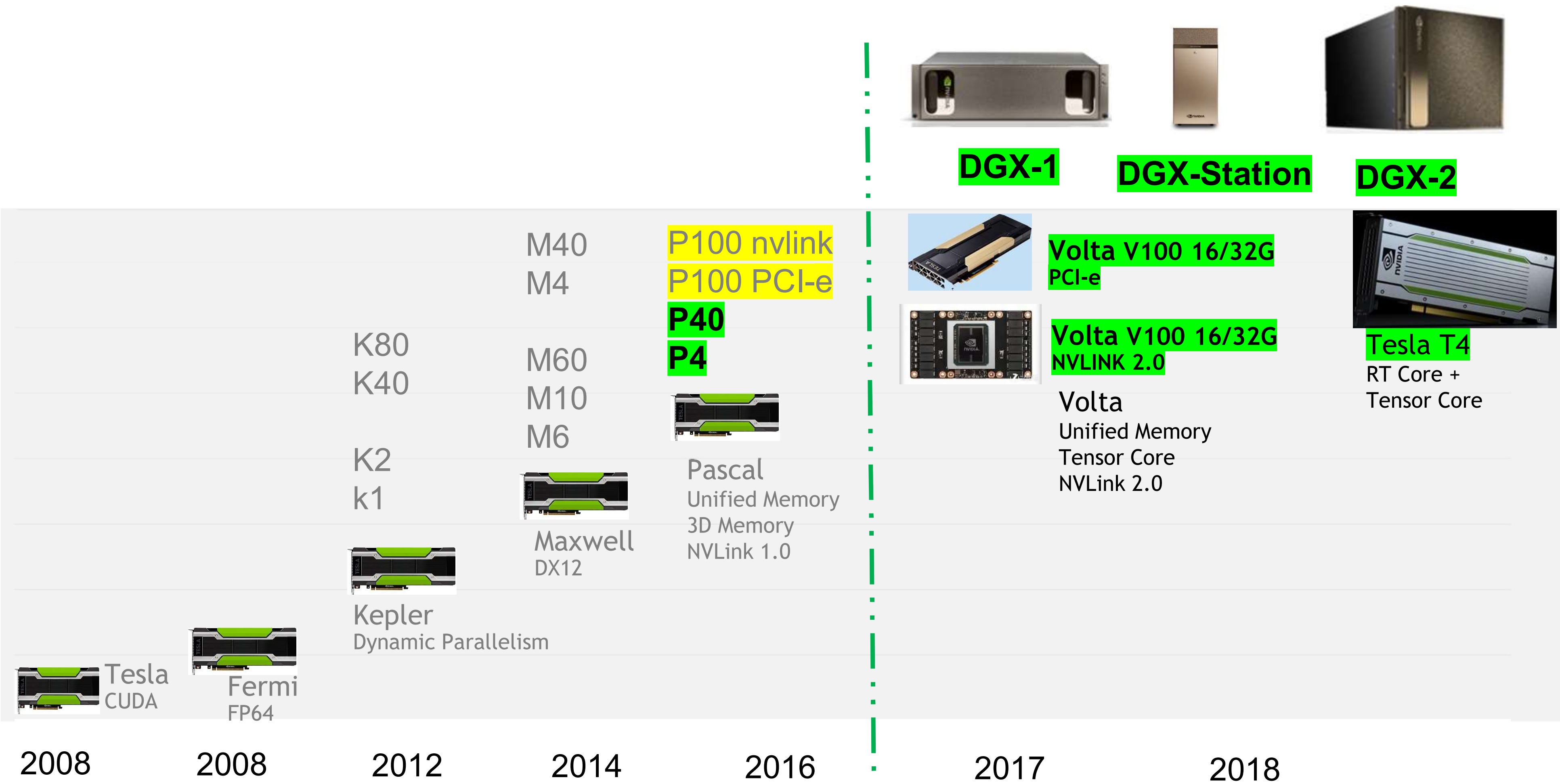
2018

- cuBLAS: 9.0
- cuFFT: 9.0
- cuRAND: 9.0
- cuSOLVER: 9.0
- cuSPARSE: 9.0
- NPP: 9.0
- Thrust: 1.9.0
- CUDA: 9.0
- Resource Mgr: r384
- Base OS: Ubuntu 16.04

Accelerated Server
with Volta

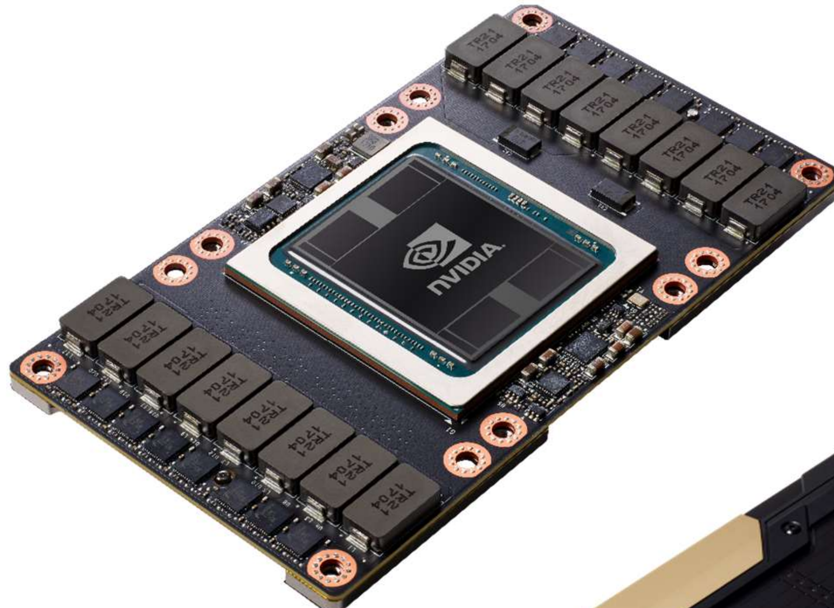
英伟达企业级GPU产品TESLA

TESLA 路线图


DGX-1
DGX-Station
DGX-2
M40
M4
P100 nvlink
P100 PCI-e
P40
P4
K80
K40
M60
M10
M6
K2
k1
Volta V100 16/32G
PCI-e
Volta V100 16/32G
NVLINK 2.0
Volta
Unified Memory
Tensor Core
NVLink 2.0
Tesla T4
RT Core +
Tensor Core
Pascal
Unified Memory
3D Memory
NVLink 1.0
Maxwell
DX12
Kepler
Dynamic Parallelism
Tesla
CUDA
Fermi
FP64
2008
2008
2012
2014
2016
2017
2018

TESLA V100 NV-LINK

5,120 CUDA cores
640 NEW Tensor cores
7.8 FP64 TFLOPS | 15.7 FP32 TFLOPS
125 Tensor TFLOPS
20MB SM RF | 16MB Cache | 32GB HBM2 @ 900 GB/s
300 GB/s NVLink

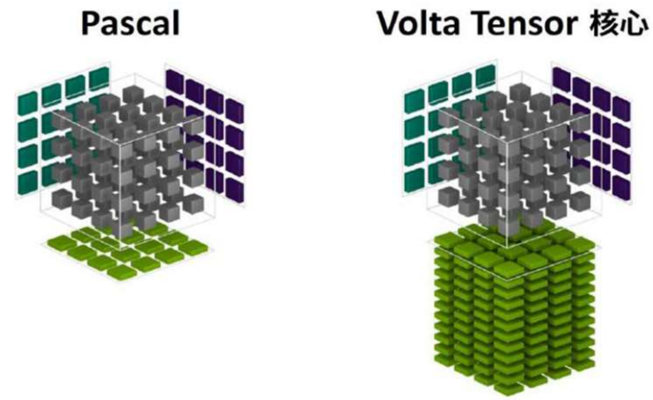


Tesla V100 PCIe

5,120 CUDA cores | 640 Tensor cores
7.0 FP64 TFLOPS | 14 FP32 TFLOPS
NEW 112 Tensor TFLOPS
20MB SM RF | 16MB Cache | 32GB 16GB HBM2 @ 900 GB/s
32 GB/s PCIe Gen3

TENSOR CORE

Mixed Precision Matrix Math
4x4 matrices



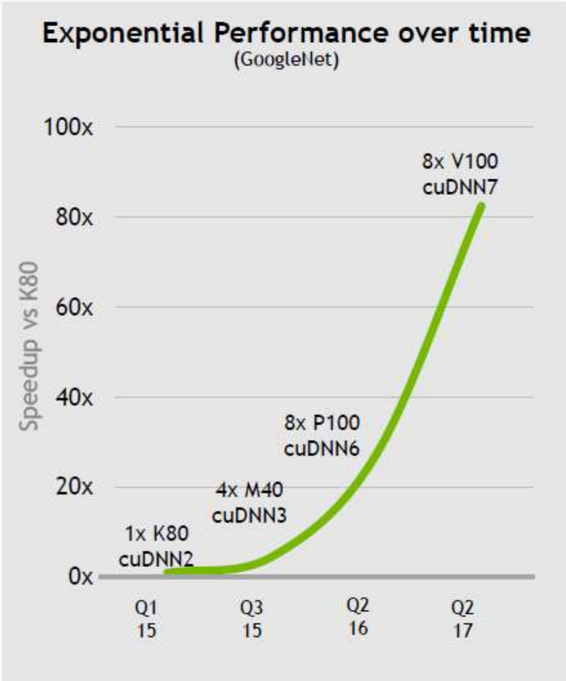
$$D = \begin{pmatrix} A_{0,0} & A_{0,1} & A_{0,2} & A_{0,3} \\ A_{1,0} & A_{1,1} & A_{1,2} & A_{1,3} \\ A_{2,0} & A_{2,1} & A_{2,2} & A_{2,3} \\ A_{3,0} & A_{3,1} & A_{3,2} & A_{3,3} \end{pmatrix} \begin{pmatrix} B_{0,0} & B_{0,1} & B_{0,2} & B_{0,3} \\ B_{1,0} & B_{1,1} & B_{1,2} & B_{1,3} \\ B_{2,0} & B_{2,1} & B_{2,2} & B_{2,3} \\ B_{3,0} & B_{3,1} & B_{3,2} & B_{3,3} \end{pmatrix} + \begin{pmatrix} C_{0,0} & C_{0,1} & C_{0,2} & C_{0,3} \\ C_{1,0} & C_{1,1} & C_{1,2} & C_{1,3} \\ C_{2,0} & C_{2,1} & C_{2,2} & C_{2,3} \\ C_{3,0} & C_{3,1} & C_{3,2} & C_{3,3} \end{pmatrix}$$

FP16 or FP32   FP16   FP16   FP16 or FP32

$$D = AB + C$$

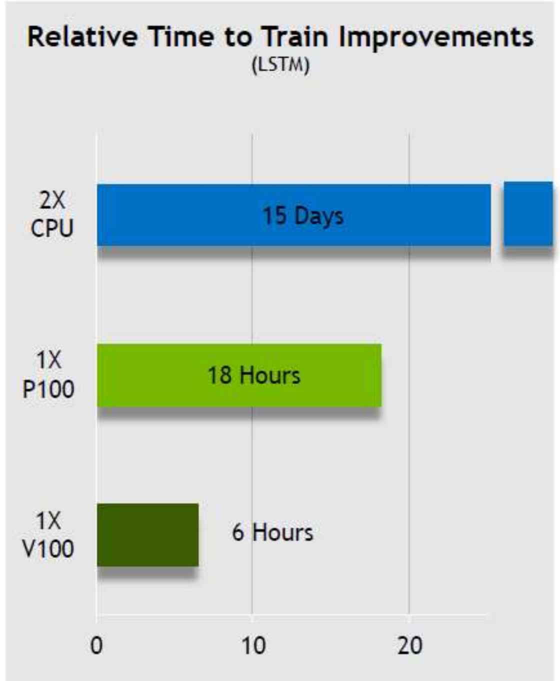
V100带来变革性的AI性能提升

3X Faster DL Training Performance



Over 80X DL Training
Performance in 3 Years

GoogleNet Training Performance on versions of cuDNN
Vs 1x K80 cuDNN2



3X Reduction in Time to Train Over P100

Neural Machine Translation Training for 13 Epochs |German ->English, WMT15 subset | CPU = 2x
Xeon E5 2699 V4

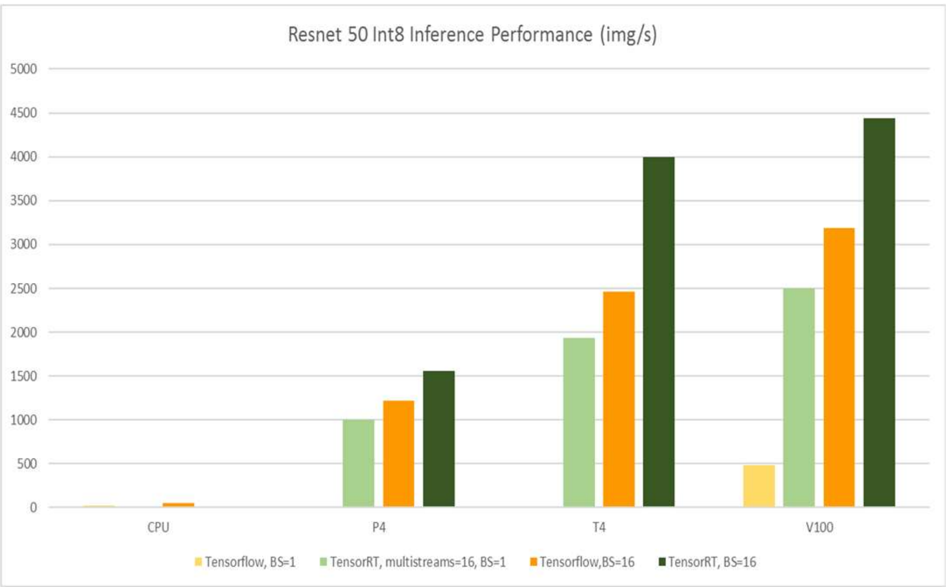
TESLA T4

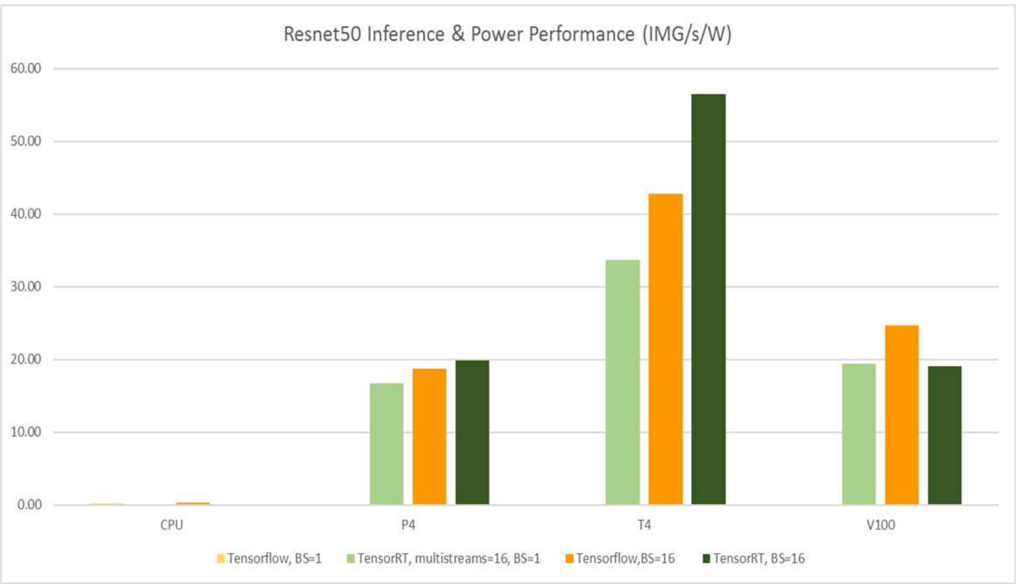
WORLD'S MOST ADVANCED INFERENCE GPU

Universal Inference Acceleration
320 Turing Tensor cores
2,560 CUDA cores
8.1 FP32 TFLOPS
65 FP16 TFLOPS | 130 INT8 TOPS | 260 INT4 TOPS
16GB | 320GB/s

>2.5X INFERENCE PERFORMANCE OF P4

Measured across frameworks and batch sizes



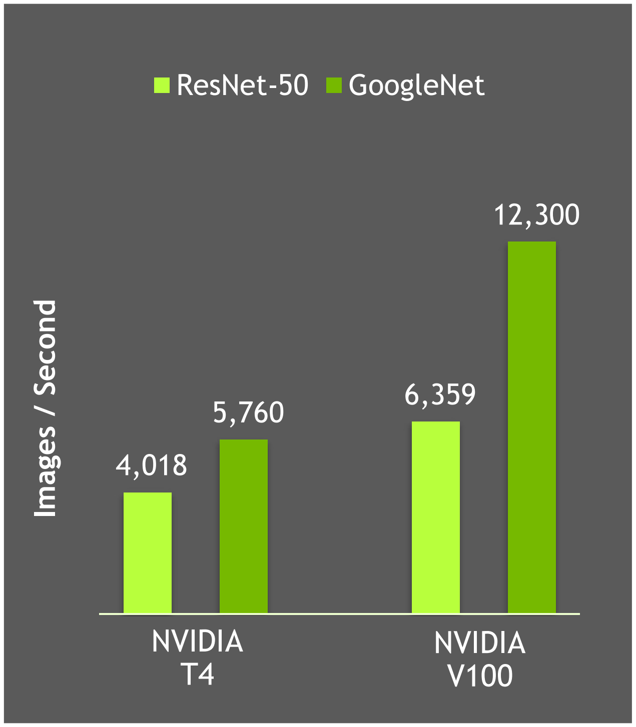


P4,T4 and V100 precision=int8. cuDNN Version=7.3.0.29. CUDA 10. NGC container 18.09. T4 software is still being optimized. Multistream, batch=1software is in development.

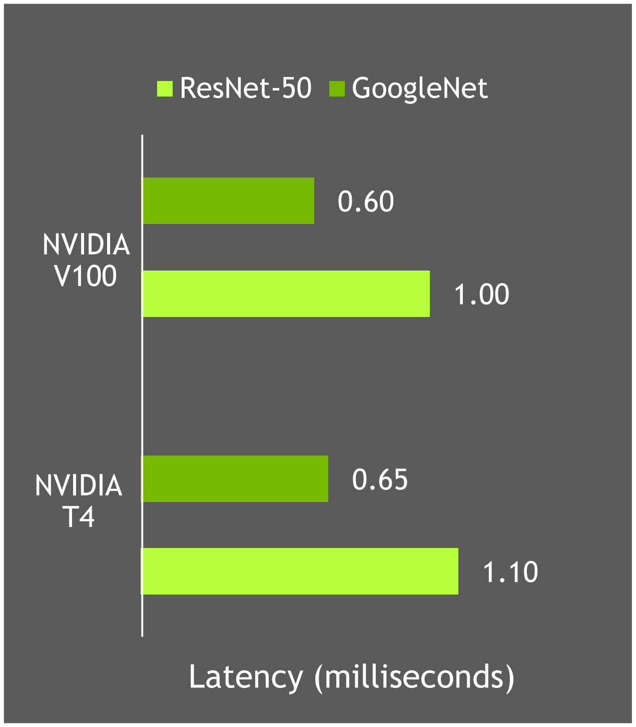
WORLD'S FASTEST INFERENCE PERFORMANCE

NVIDIA GPUs Set New Performance Records

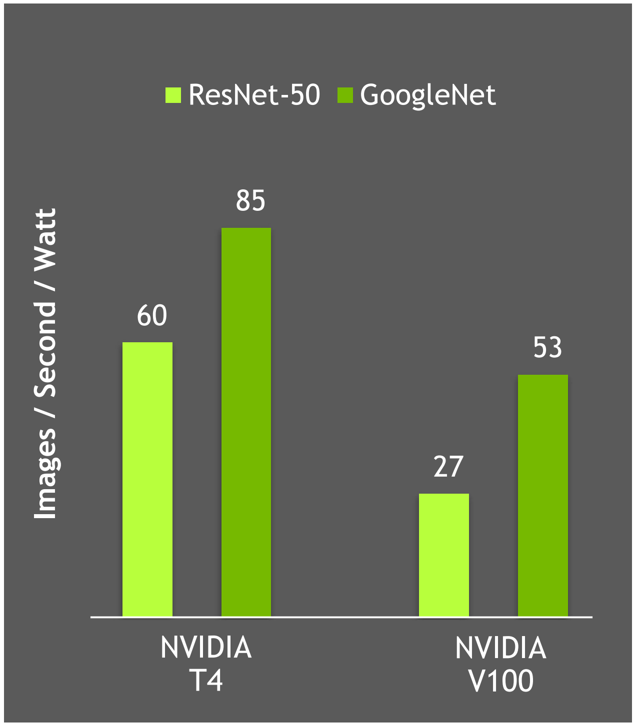
Throughput



Latency



Energy Efficiency



AI超级计算机DGX系统

世界第一的个人 AI 超级计算机
NVIDIA DGX STATION

超强计算能力，液冷无声

480 Tensor TFLOPS | 4x Tesla V100 16GB

NVLink Fully Connected | 3x DisplayPort

1500W | Water Cooled

[redacted]

NVIDIA DGX-1
世界第一台深度学习超级计算机
计算能力约为400台CPU服务器

1 PFLOPS 深度学习计算能力

8张Tesla V100 GPU

NVLink 300GB/s GPU高速互联

天然支持所有的深度学习框架

双至强CPU

7 TB固态硬盘

双10Gb以太网

4个100Gb Infiniband网络

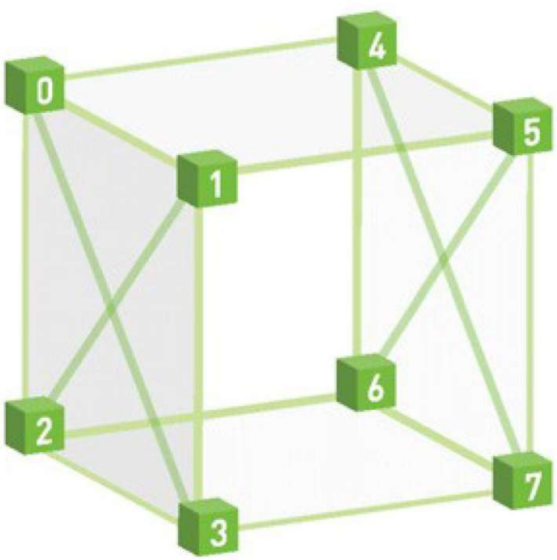
3RU – 3200W

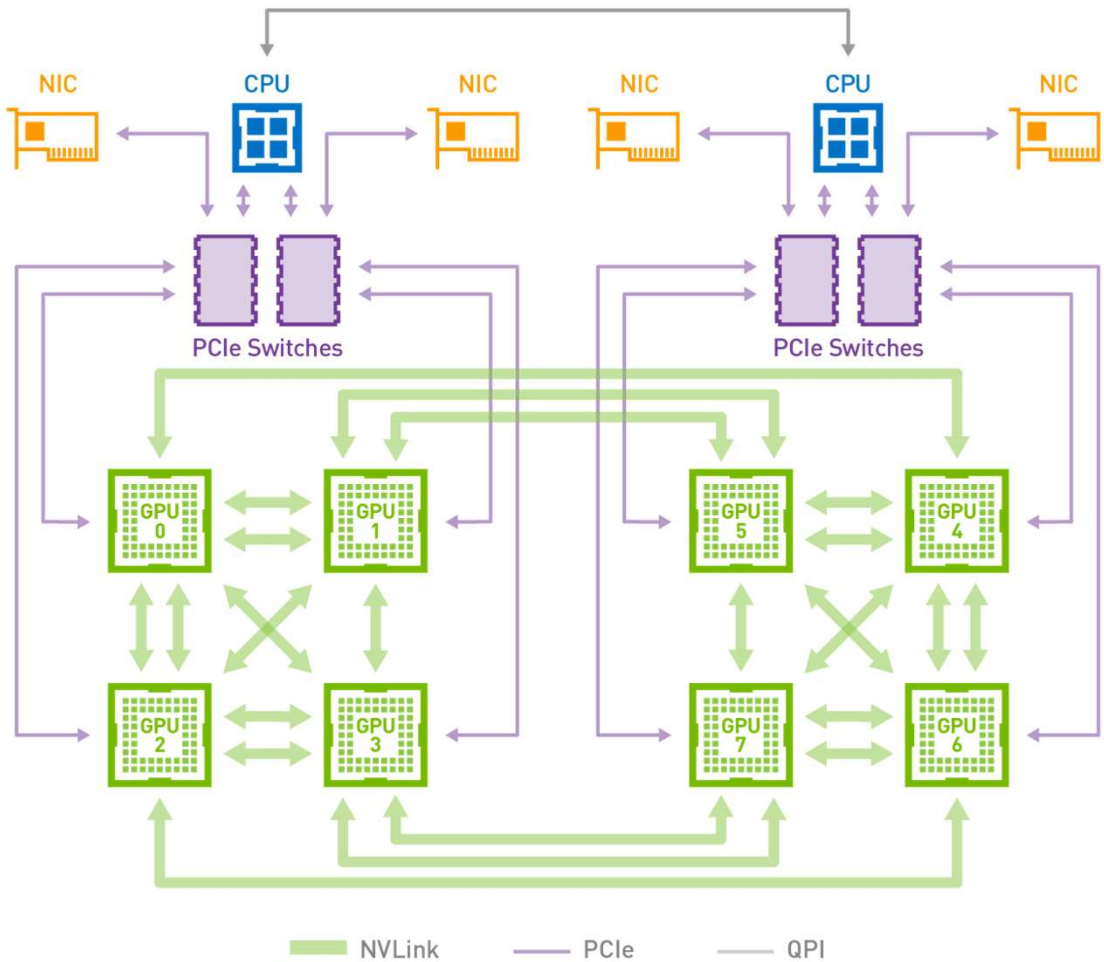
VOLTA NVLINK

300GB/sec

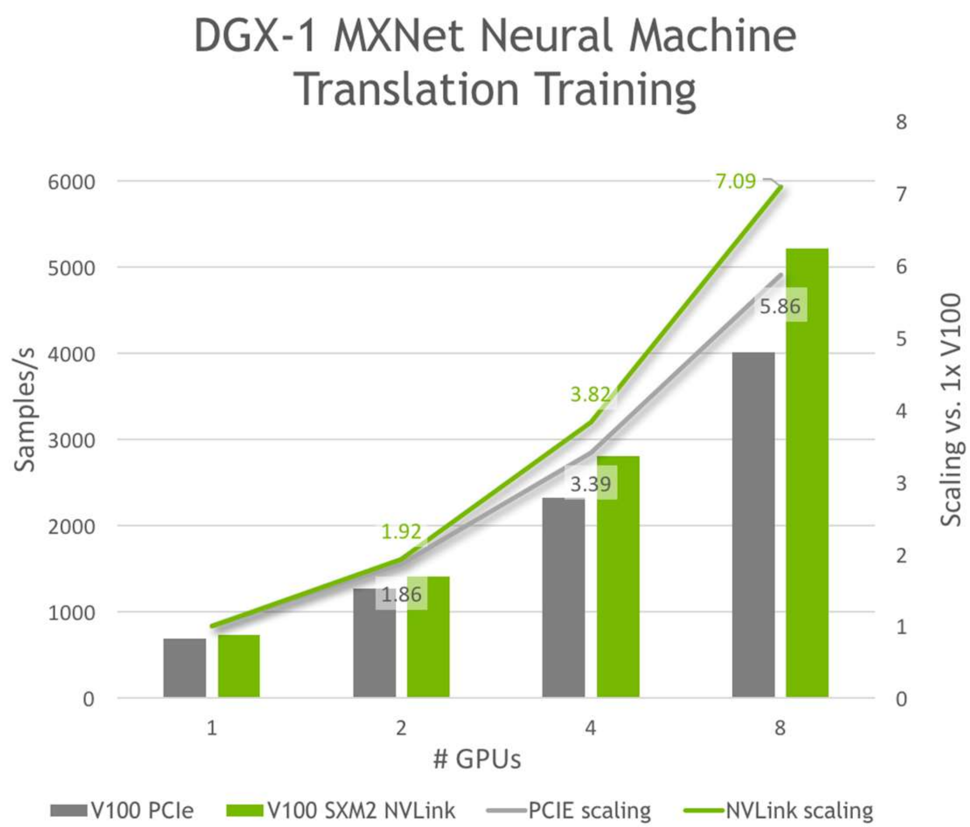
50% more links

28% faster signaling





30% BETTER PERFORMANCE WITH NVLINK THAN PCIE



- Encoder and decoder embedding size of 512
- Batch size of 256 per GPU
- NVIDIA DGX containers version 17.11, processing real data with cuDNN 7.0.4, NCCL 2.1.2

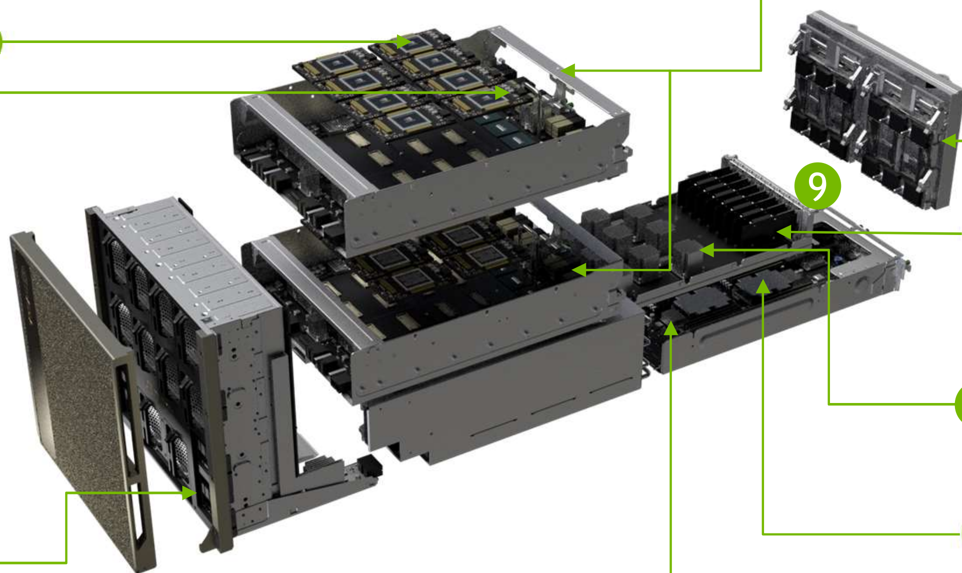
NVIDIA DGX-2世界最强的超级计算机

THE WORLD'S MOST POWERFUL DEEP LEARNING SYSTEM FOR THE MOST COMPLEX DEEP LEARNING CHALLENGES

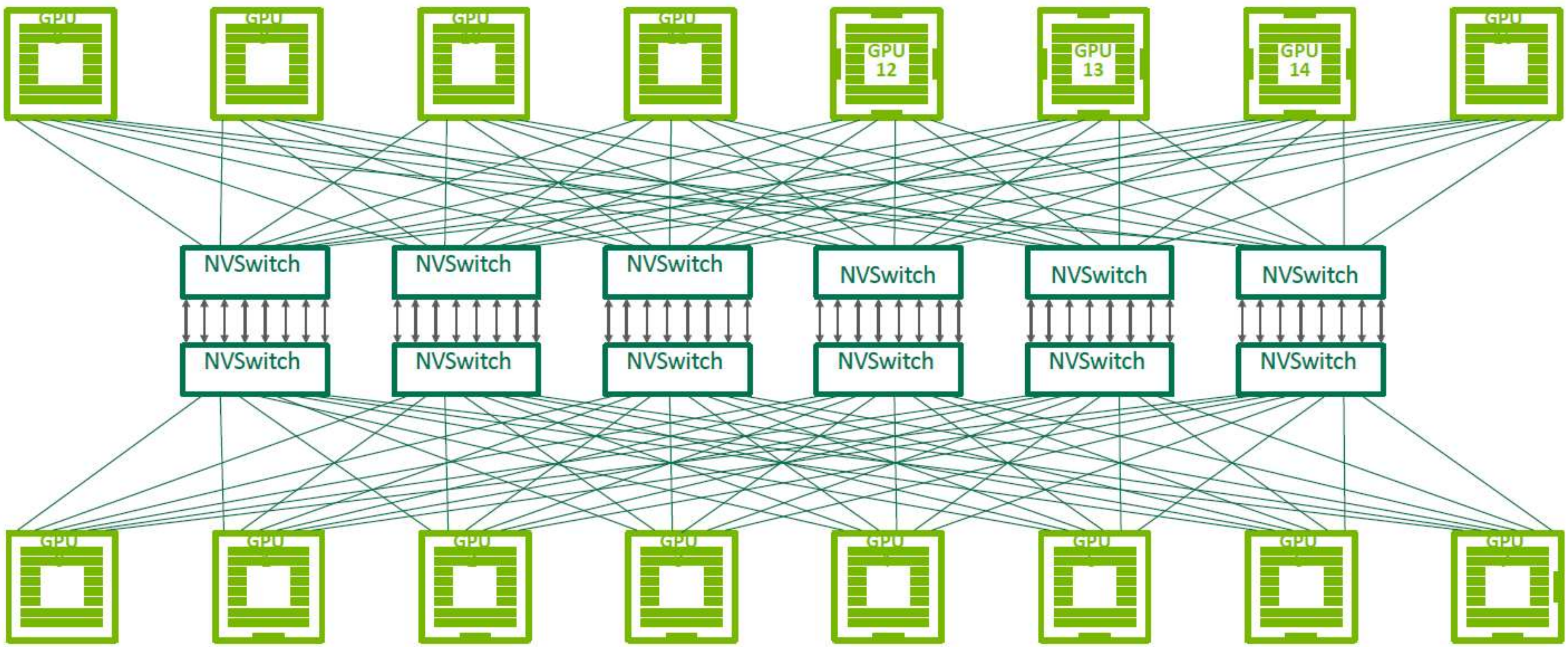
- First 2 PFLOPS System
- 16 V100 32GB GPUs Fully Interconnected
- NVSwitch: 2.4 TB/s bisection bandwidth
- 24X GPU-GPU Bandwidth
- 0.5 TB of Unified GPU Memory
- 10X Deep Learning Performance

Support in CUDA 10.0

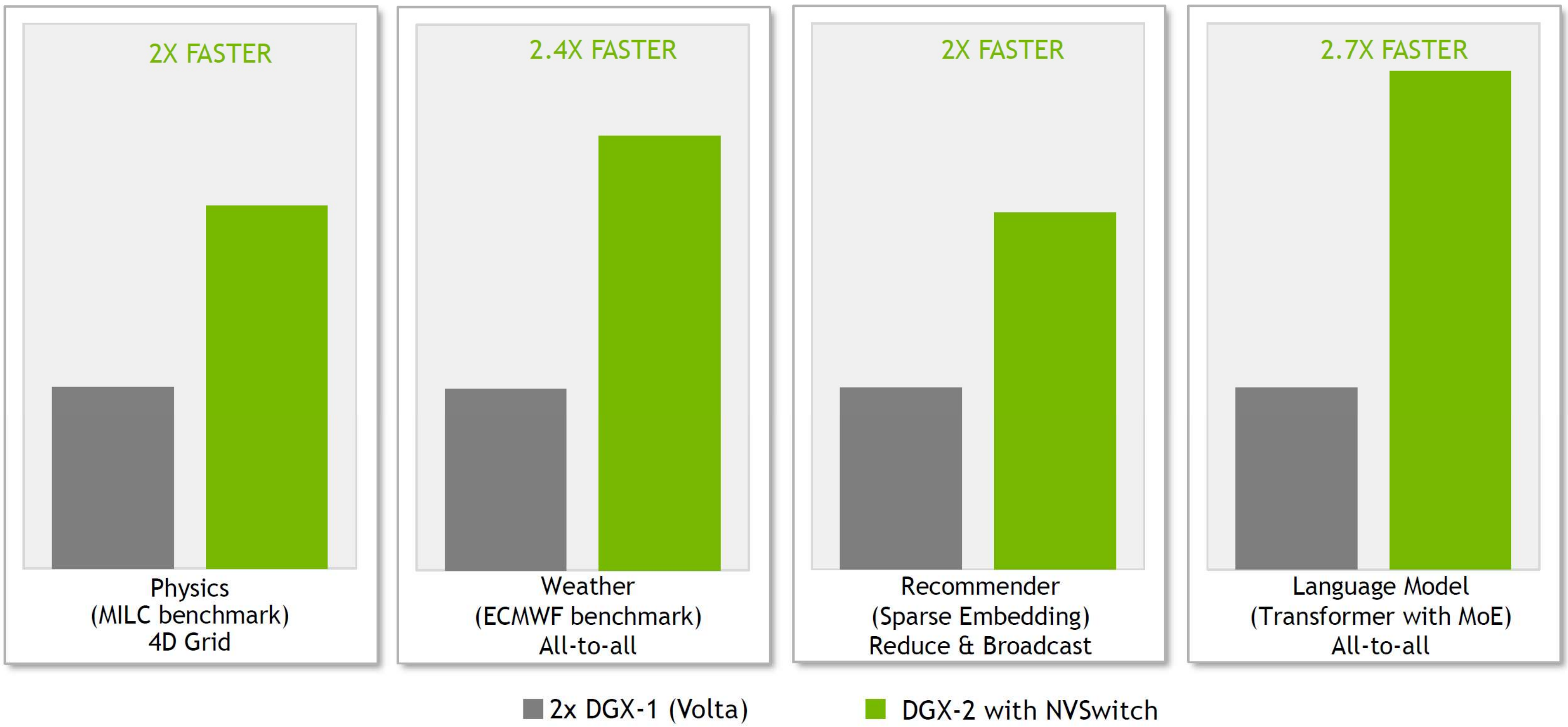
DESIGNED TO TRAIN THE PREVIOUSLY IMPOSSIBLE



FULL NON-BLOCKING BANDWIDTH



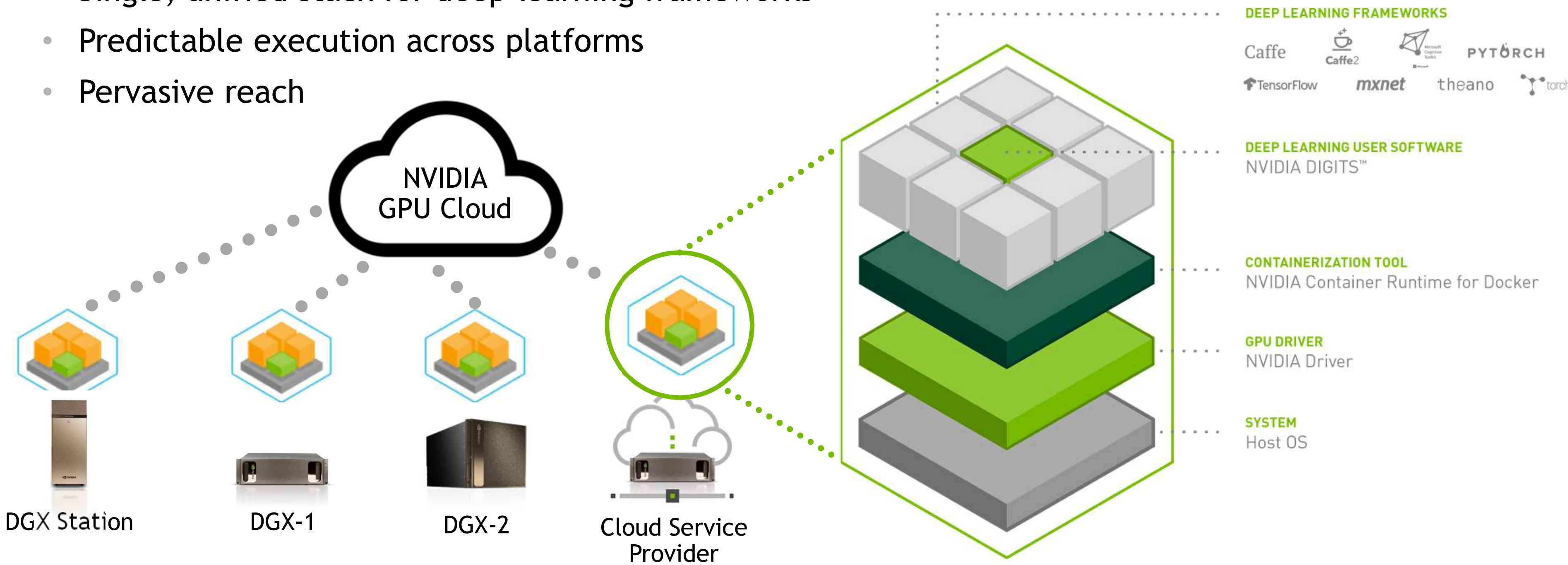
2X HIGHER PERFORMANCE WITH NVSWITCH



2 DGX-1V servers have dual socket Xeon E5 2698v4 Processor. 8 x V100 GPUs. Servers connected via 4X 100Gb IB ports | DGX-2 server has dual-socket Xeon Platinum 8168 Processor. 16 V100 GPUs

COMMON SOFTWARE STACK ACROSS DGX FAMILY

- Single, unified stack for deep learning frameworks
- Predictable execution across platforms
- Pervasive reach



NGC CONTAINER REGISTRY

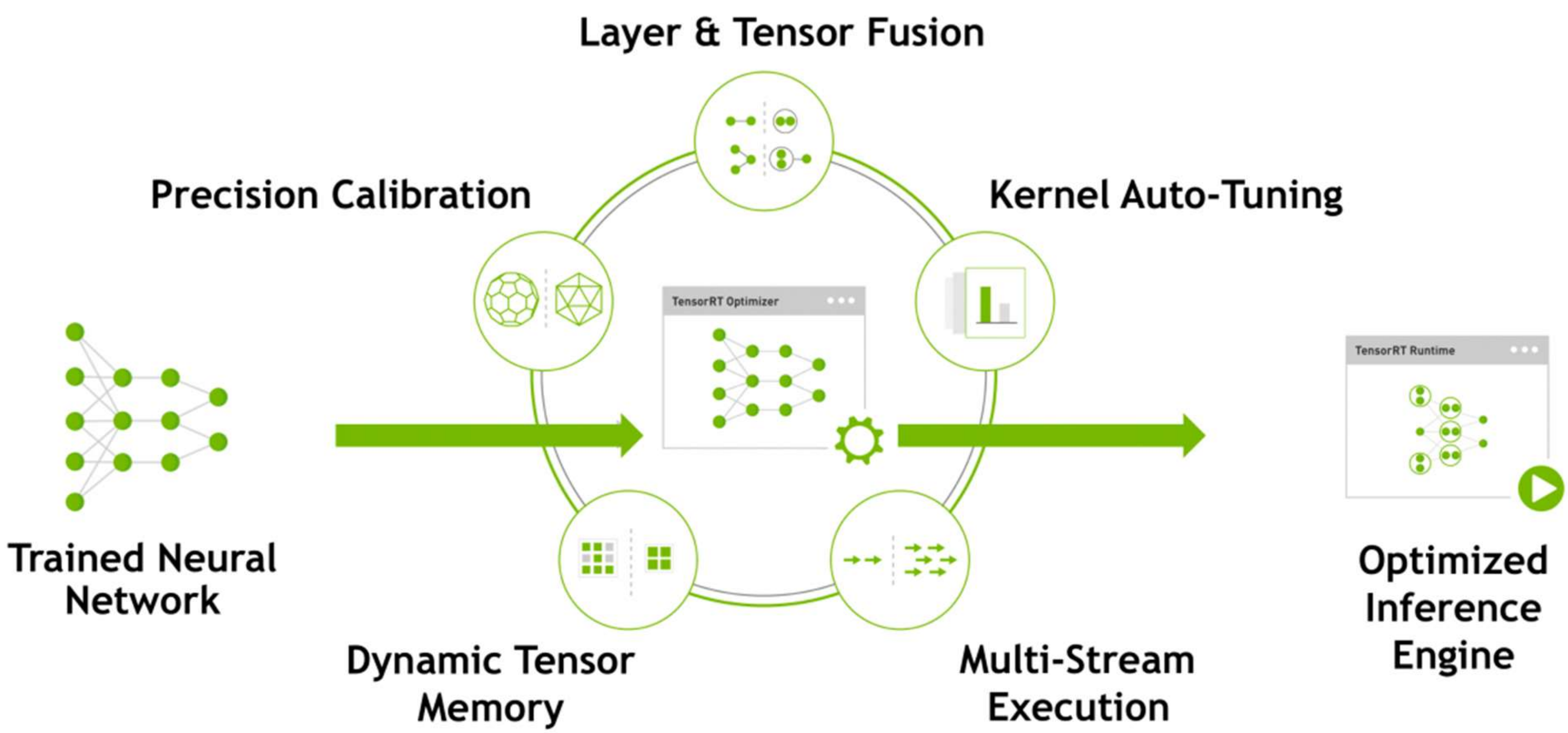
10 Containers at Launch, 35 Containers Today

Deep Learning	HPC	HPC Visualization	NVIDIA/K8s	Partners
caffe	bigdft	index	Kubernetes on NVIDIA GPUs	chainer
caffe2	candle	paraview-holodeck		h20ai-driverless
cntk	chroma	paraview-index		kinetica
cuda	gamess	paraview-optix		mapd
digits	gromacs			paddlepaddle
inferenceserver	lammps			
mxnet	lattice-microbes			
pytorch	milc			
tensorflow	namd			
tensorrt	pgi			
theano	picongpu			
torch	relion			
	vmd			

NVIDIA GPU CLOUD

ngc.nvidia.com

TENSORRT OPTIMIZER



实际的性能提升

GLOBAL TECHNOLOGY FIRM SPECIALIZING IN DIGITAL MEDIA

Home-grown "optimized" TensorFlow stack

DGX+NGC TensorFlow stack

1.5X FASTER

1680 images/sec

2600 images/sec

ResNet50 Training

WORLD-LEADING MEDICAL RESEARCH CENTER

Home-grown TensorFlow stack

DGX+NGC TensorFlow stack

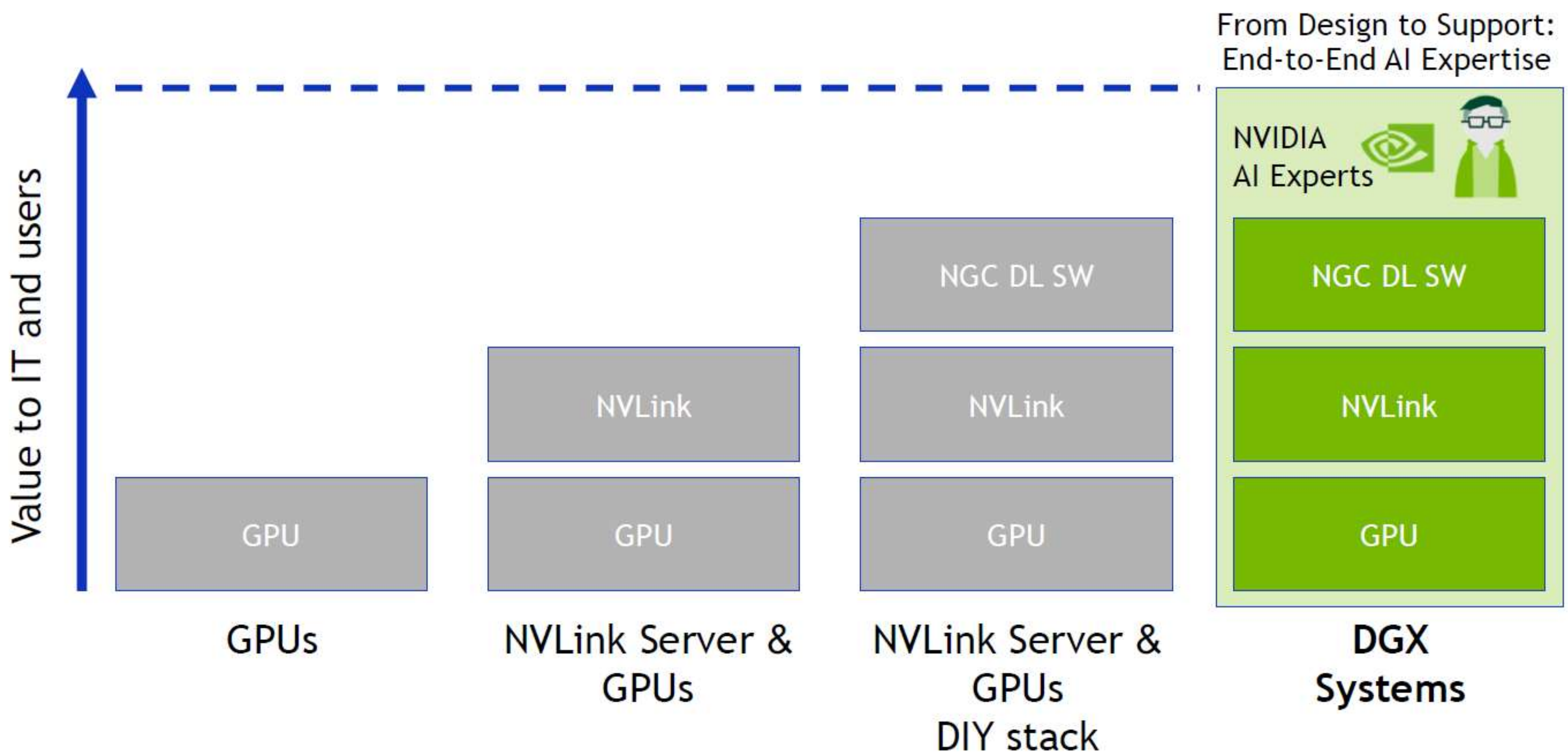
2.1X FASTER

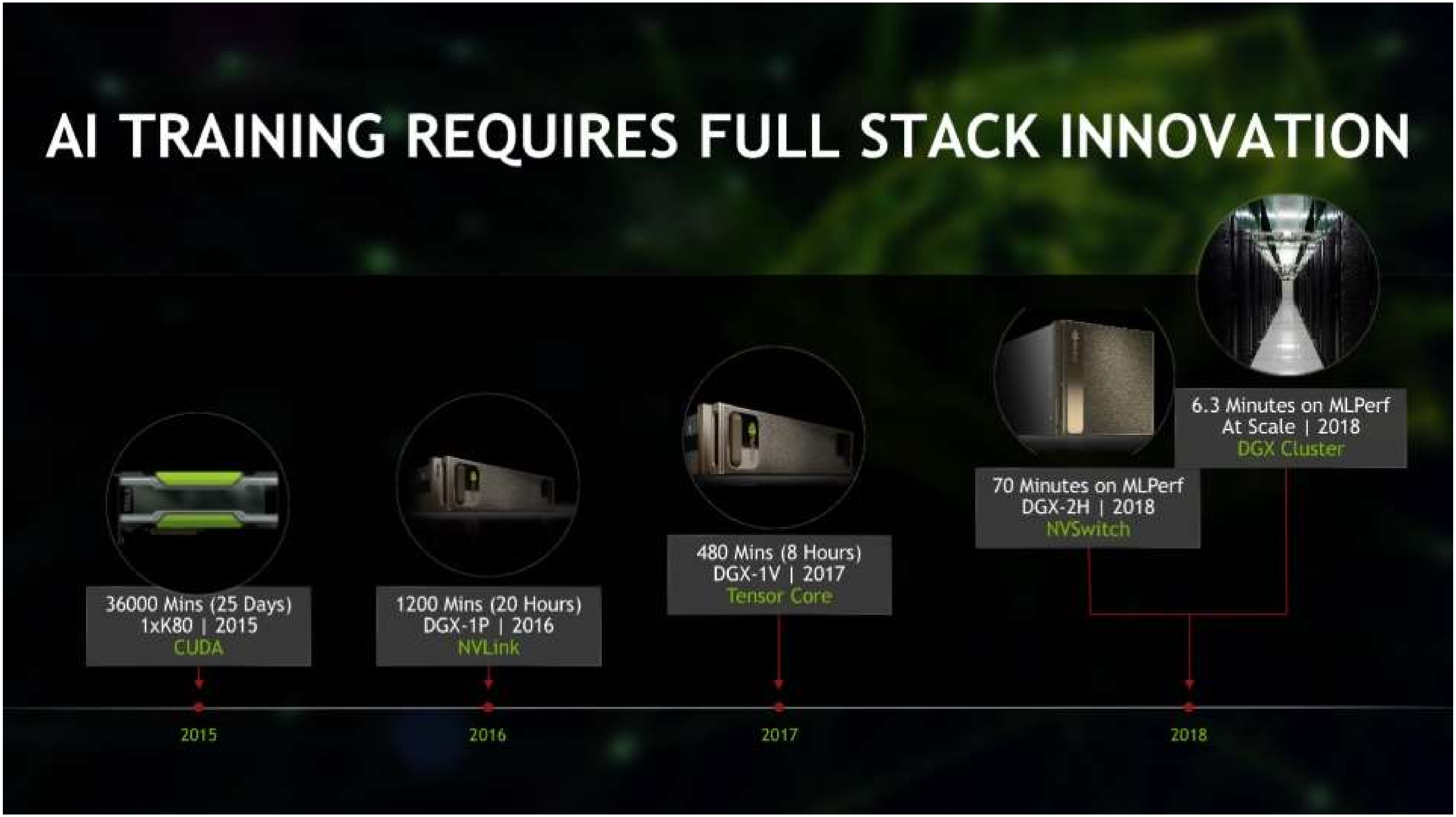
1238 images/sec

2600 images/sec

ResNet50 Training

DGX SYSTEMS — THE VALUE OF INTEGRATED SOLUTIONS AND AI EXPERTISE



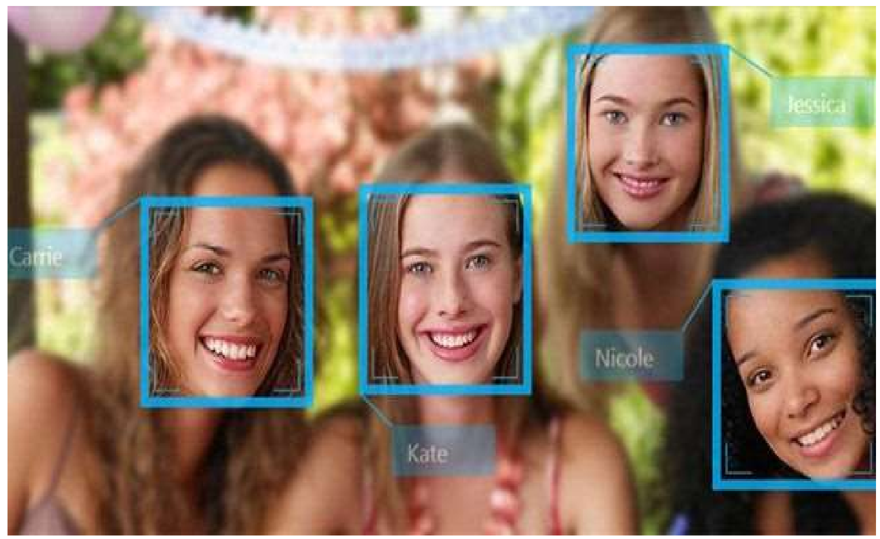

AI TRAINING REQUIRES FULL STACK INNOVATION
36000 Mins (25 Days)
1xK80 | 2015
CUDA
1200 Mins (20 Hours)
DGX-1P | 2016
NVLink
480 Mins (8 Hours)
DGX-1V | 2017
Tensor Core
70 Minutes on MLPerf
DGX-2H | 2018
NVSwitch
6.3 Minutes on MLPerf
At Scale | 2018
DGX Cluster
2015
2016
2017
2018

TESLA/DGX 加速深度学习应用

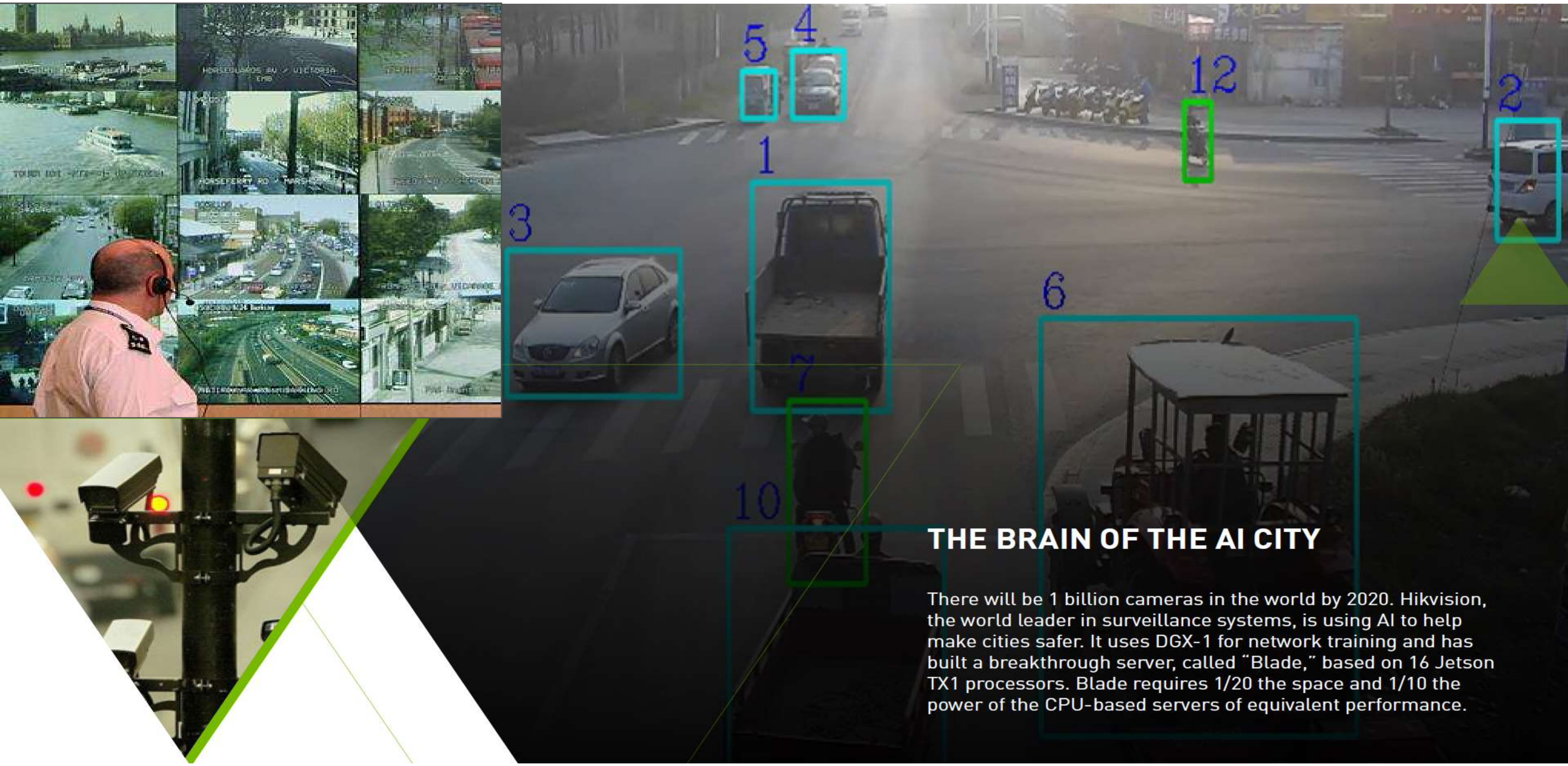
互联网行业应用


AIR FLOW


facebook
Deep Text
人工智能引擎


Jessica
Carrie
Nicole
Kate

公安行业应用

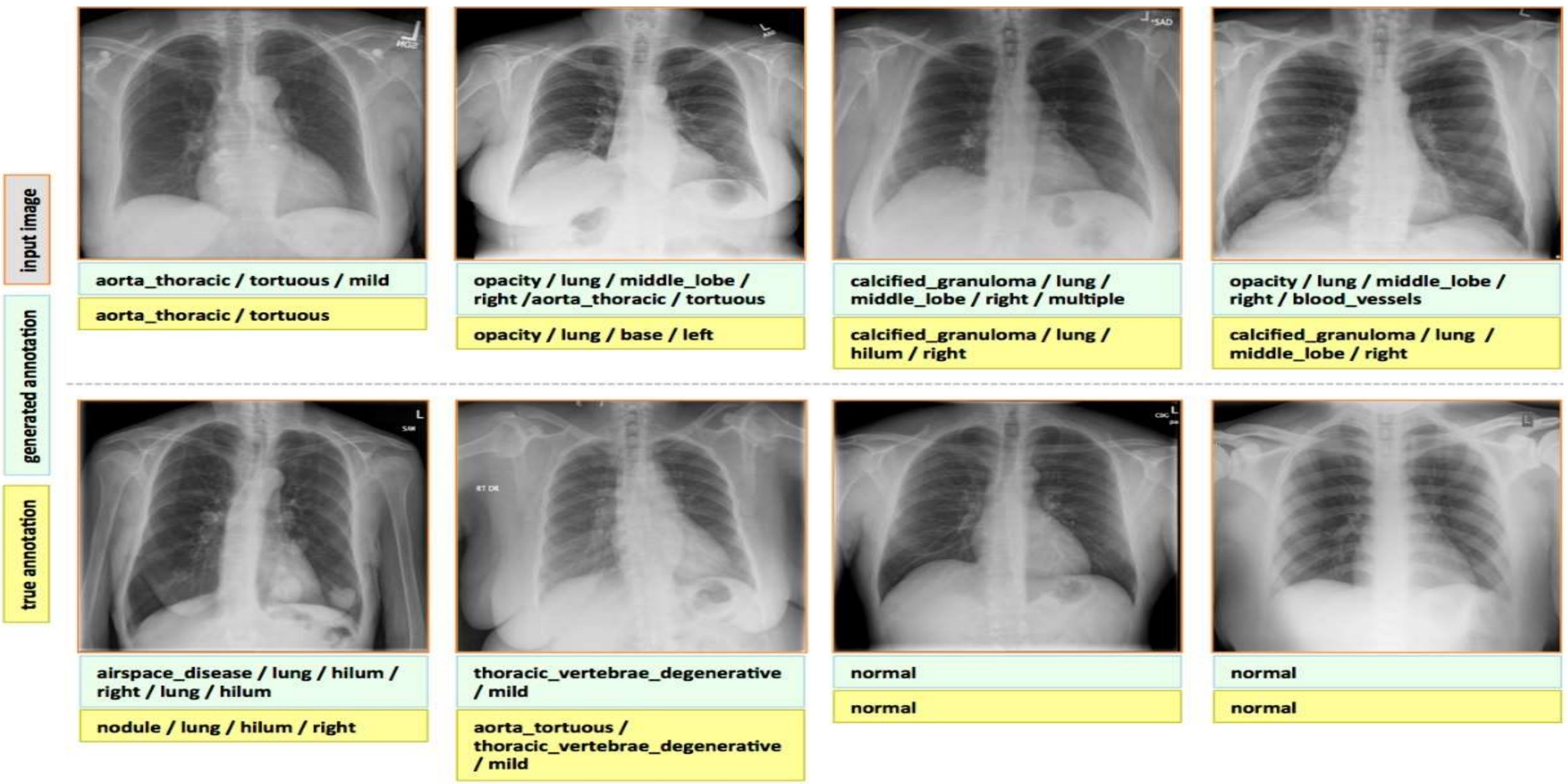

THE BRAIN OF THE AI CITY
There will be 1 billion cameras in the world by 2020. Hikvision, the world leader in surveillance systems, is using AI to help make cities safer. It uses DGX-1 for network training and has built a breakthrough server, called "Blade," based on 16 Jetson TX1 processors. Blade requires 1/20 the space and 1/10 the power of the CPU-based servers of equivalent performance.

制造行业应用

TCL-China Star Optoelectronics Technology |
IBM Visual Insights

- 150多名检查工作人员
- 平均检查时间20-30分钟
- DL（Visual Insights）<4分钟
- 与人类检查员一样准确

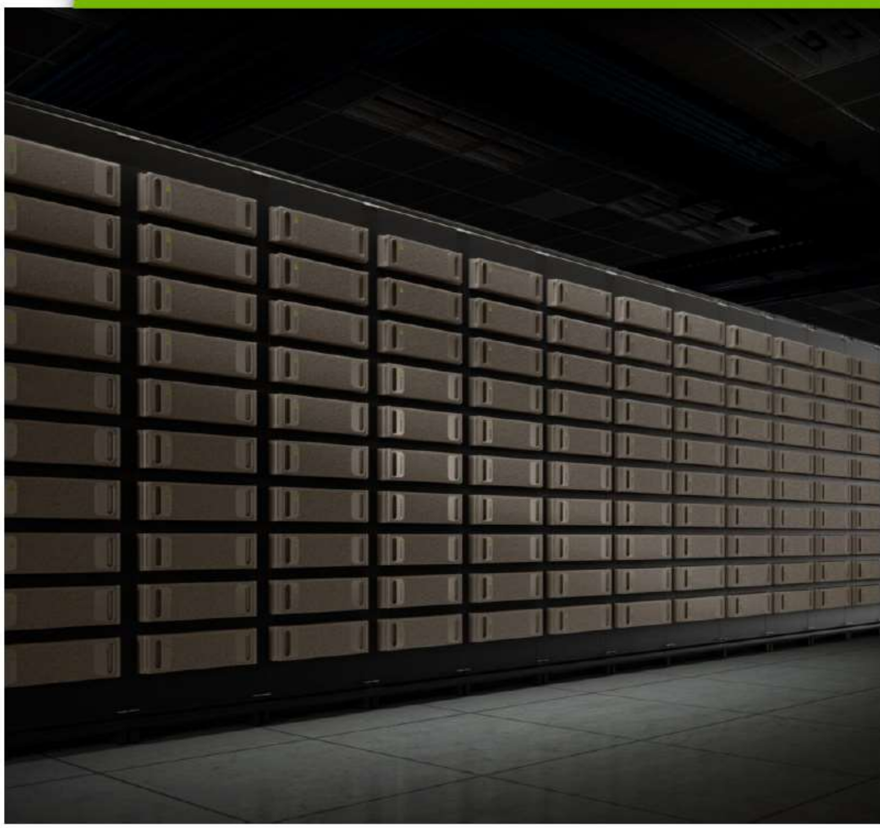
医疗影像行业的应用



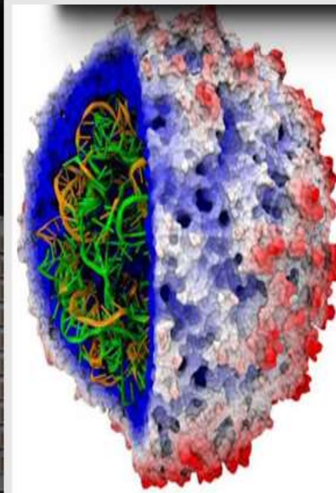
In what may be a first for radiology images, NIH researchers trained a recurrent neural network to describe the context of diseases using chest X-rays.

INTRODUCING DGX SATURNV

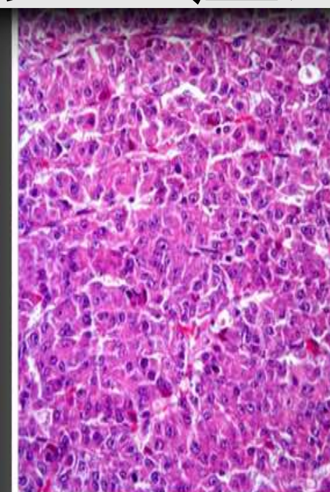
124 NVIDIA DGX-1 “Rocket for Cancer Moonshot”



癌症研究的应用



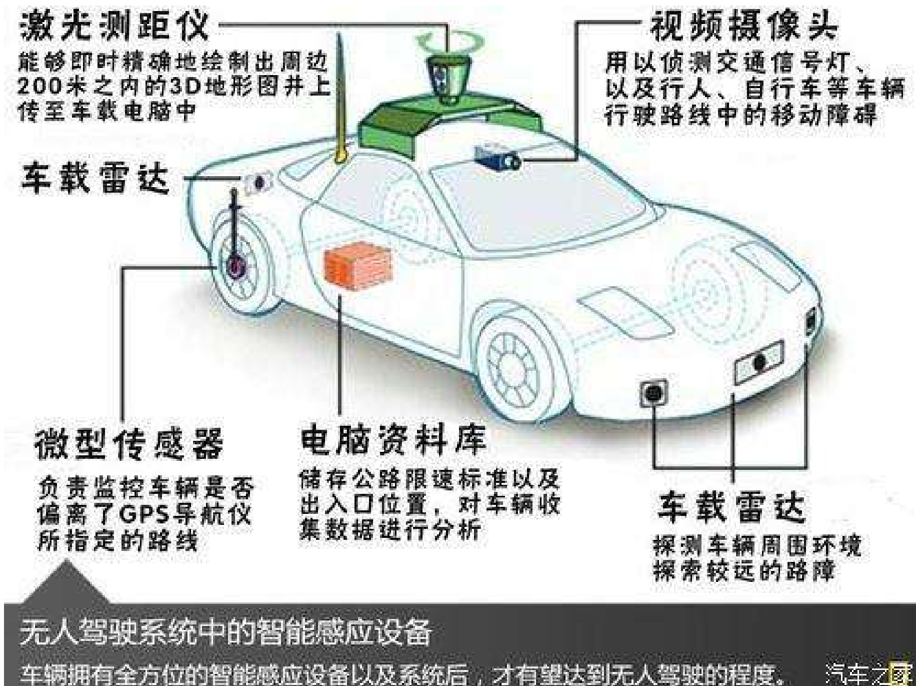
Accelerate Discovery
of Cancer Therapies

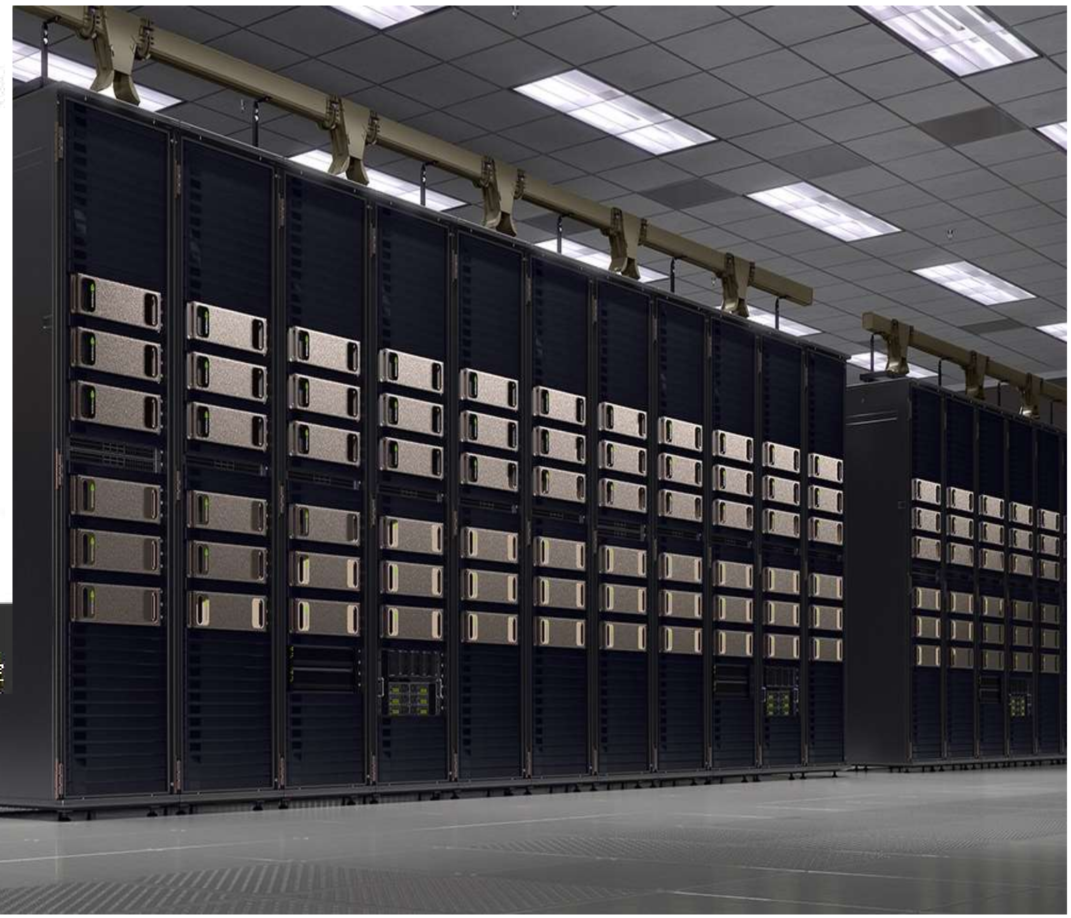


Predict Drug Response
of Cancer Patients

Automate Analysis
of Treatment Effectiveness

汽车行业应用

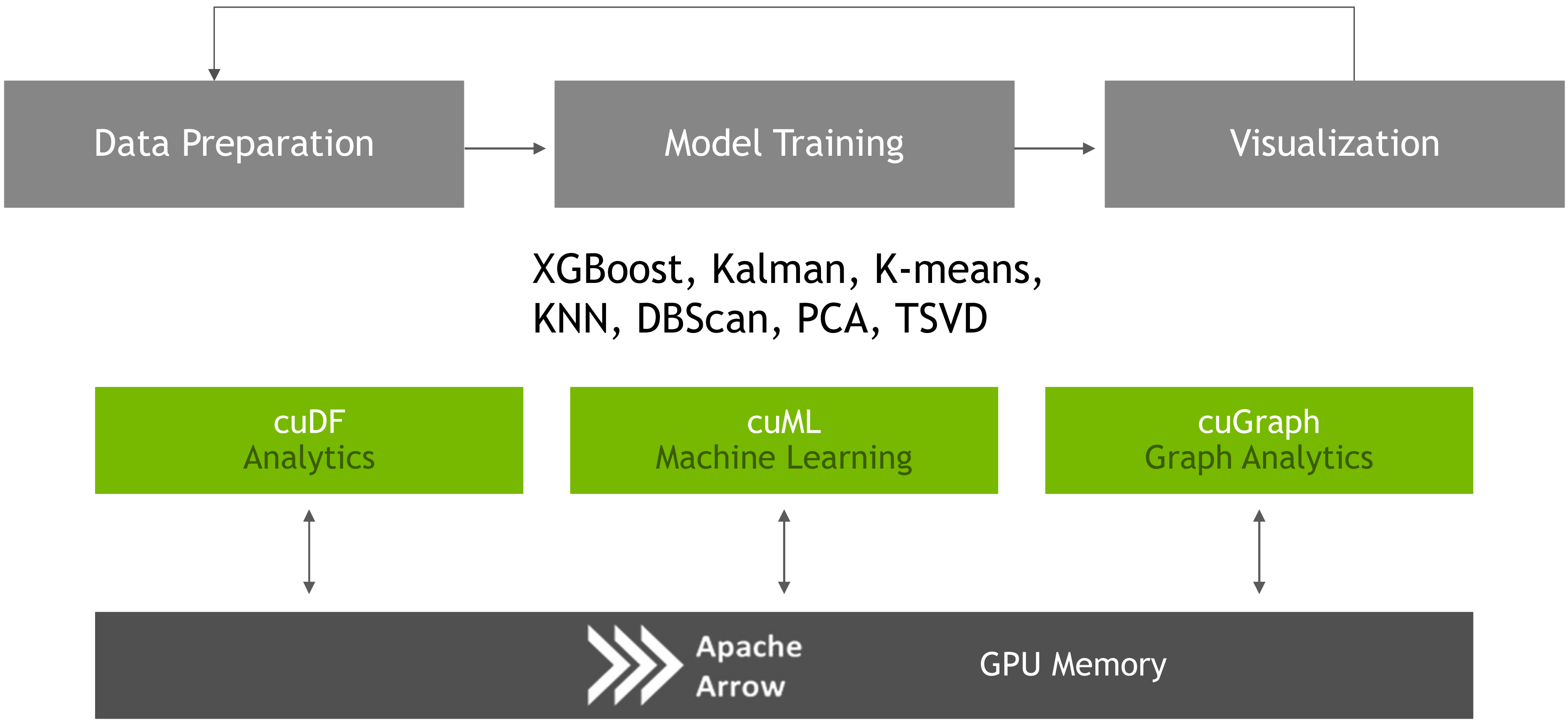

激光测距仪
能够即时精确地绘制出周边
200米之内的3D地形图并上
传至车载电脑中
视频摄像头
用以侦测交通信号灯、
以及行人、自行车等车辆
行驶路线中的移动障碍
车载雷达
微型传感器
负责监控车辆是否
偏离了GPS导航仪
所指定的路线
电脑资料库
储存公路限速标准以及
出入口位置，对车辆收
集数据进行分析
车载雷达
探测车辆周围环境
探索较远的路障
无人驾驶系统中的智能感应设备
车辆拥有全方位的智能感应设备以及系统后，才有望达到无人驾驶的程度。



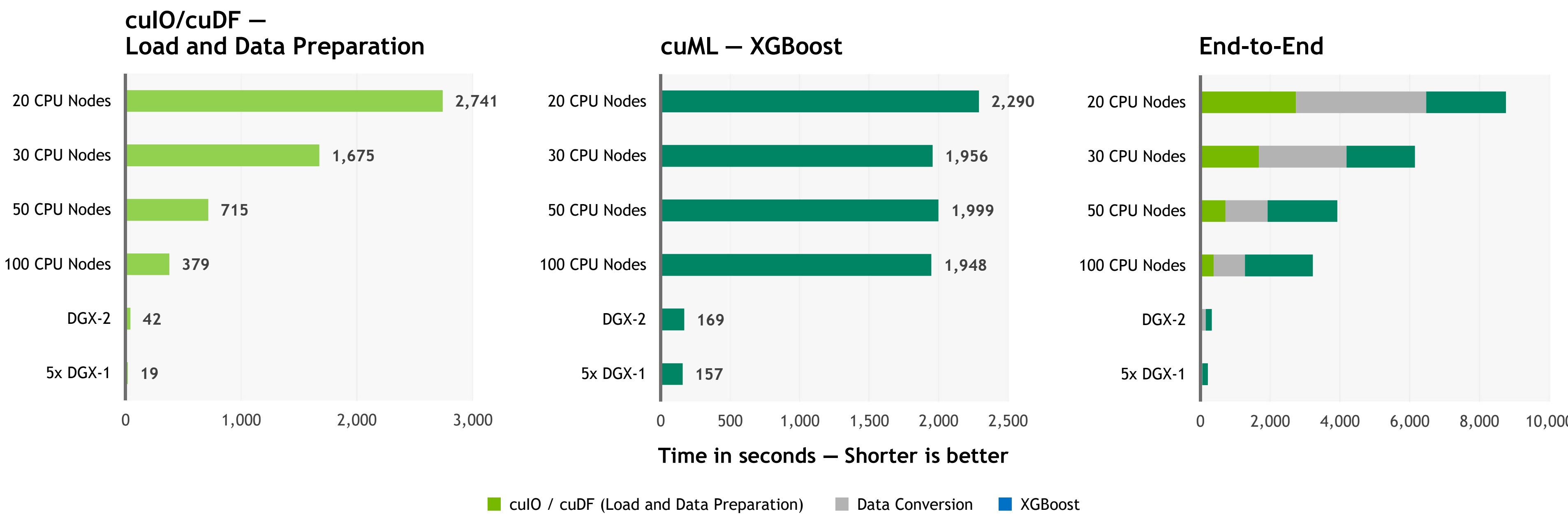
miracle of deep learning showing flowers
using resident 1:52

TESLA/DGX 加速机器学习应用

RAPIDS 开源系统加速机器学习



FASTER SPEEDS, REAL WORLD BENEFITS



Benchmark

200GB CSV dataset; Data preparation includes joins, variable transformations.

CPU Cluster Configuration

CPU nodes (61 GiB of memory, 8 vCPUs, 64-bit platform), Apache Spark

DGX Cluster Configuration

5x DGX-1 on InfiniBand network

TESLA/DGX 加速HPC应用

美国超算第一背后：英伟达GPU提供了95%计算力

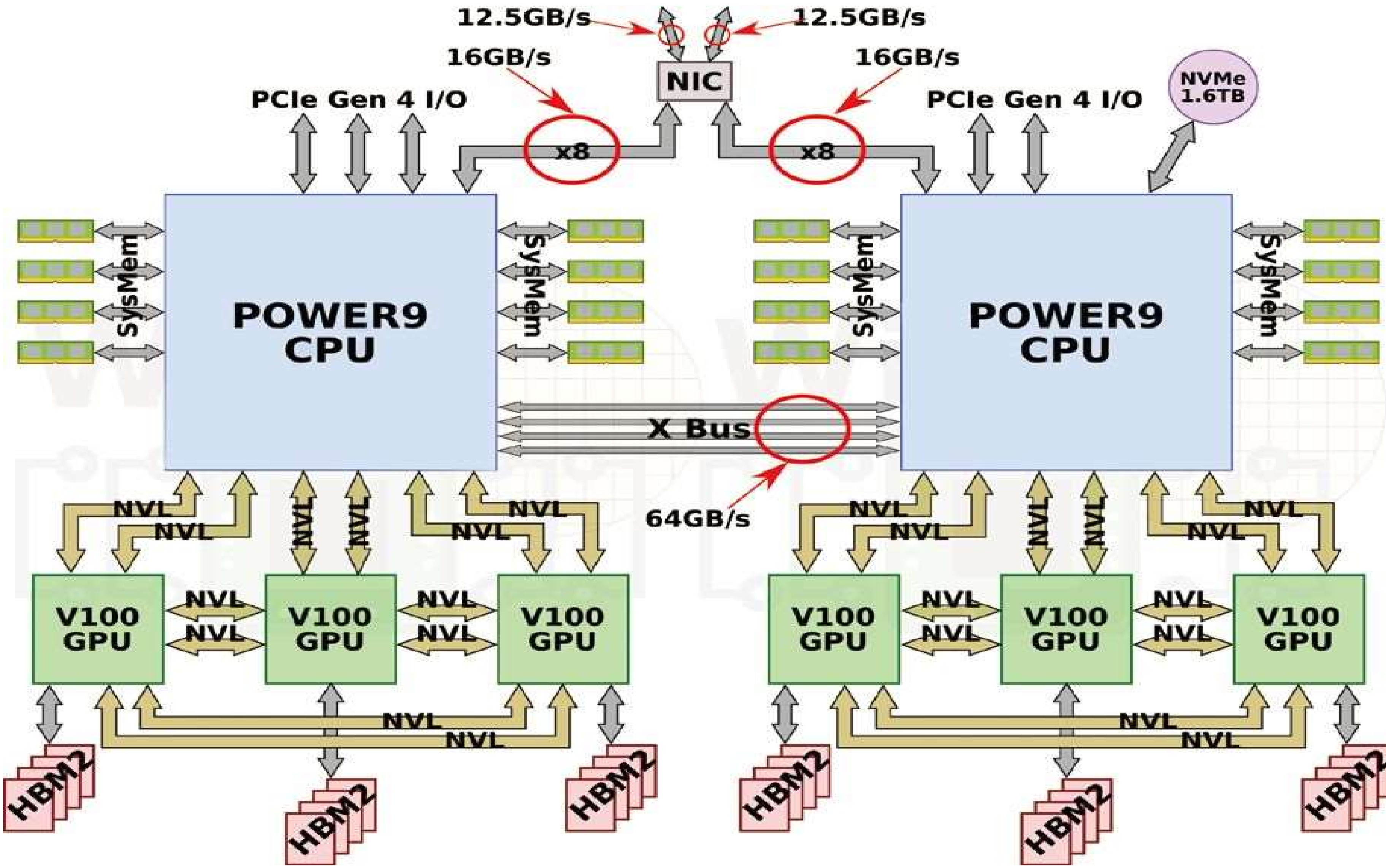
Application Performance	200 PF
Number of Nodes	4,608
Node performance	42 TF
Memory per Node	512 GB DDR4 + 96 GB HBM2
NV memory per Node	1600 GB
Total System Memory	>10 PB DDR4 + HBM2 + Non-volatile
Processors	2 IBM POWER9™ 9,216 CPUs 6 NVIDIA Volta™ 27,648 GPUs
File System	250 PB, 2.5 TB/s, GPFS™
Power Consumption	13 MW
Interconnect	Mellanox EDR 100G InfiniBand
Operating System	Red Hat Enterprise Linux (RHEL) version 7.4

Summit的性能

Summit峰值性能		
处理器	CPU	GPU
型号	POWER9	V100
数量	9,216 /2 × 18 × 256	27,648/ 6 × 18 ×256
峰值FLOPS	9.96 PF	215.7 PF
峰值AI FLOPS	N/A	3.456 EF

Summit的系统组成

Summit			
机架	计算节点	存储节点	交换机
类型	AC922	SSC (4 ESS GL4)	Mellanox IB EDR
数量	256 Racks × 18 Nodes	40 Racks × 8 Servers	18 Racks
功耗	59 kW	38 kW	N/A

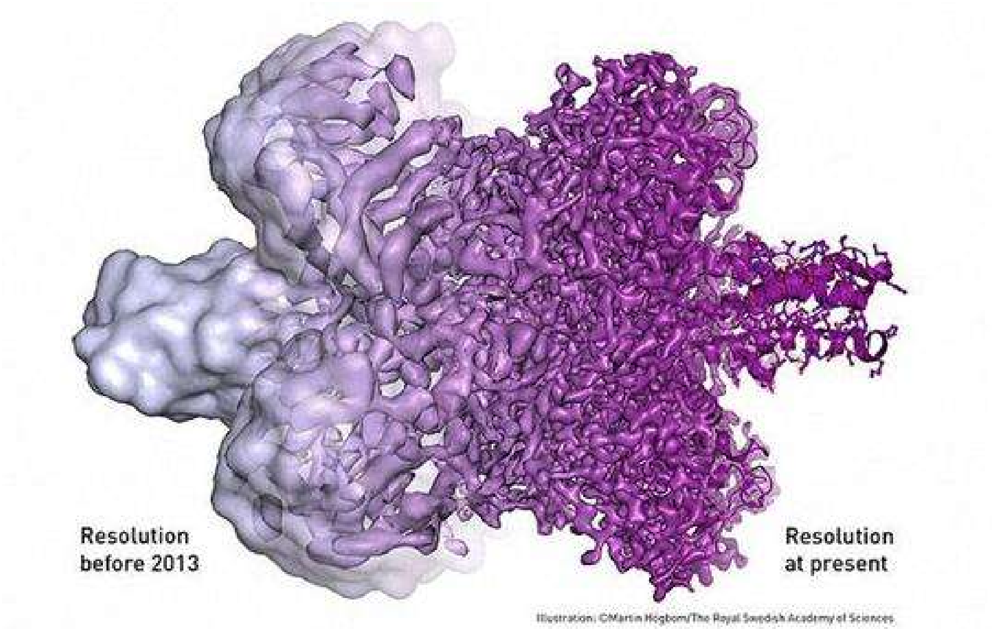


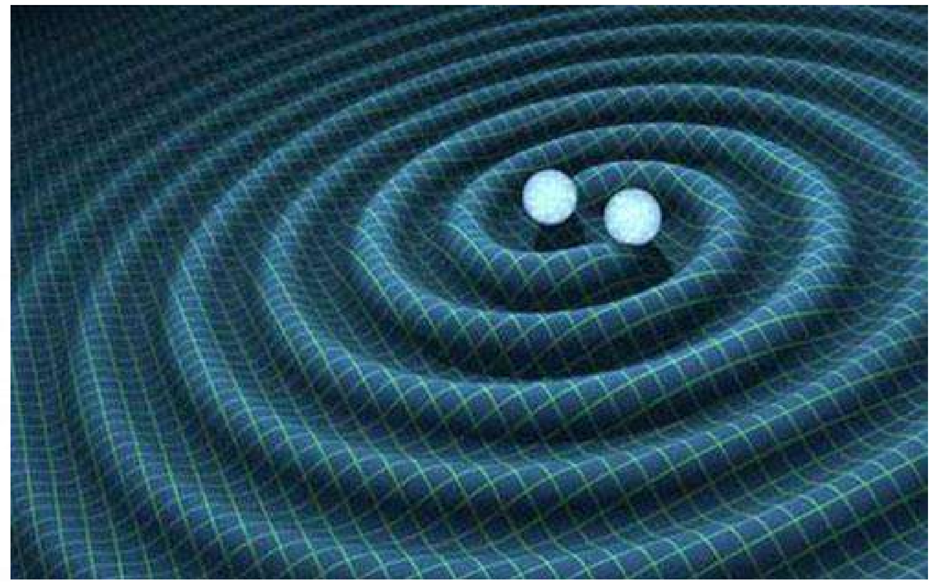
12.5GB/s
12.5GB/s
16GB/s
NIC
16GB/s
PCIe Gen 4 I/O
PCIe Gen 4 I/O
NVMe 1.6TB
x8
x8
SysMem
SysMem
SysMem
SysMem
POWER9 CPU
POWER9 CPU
X Bus
64GB/s
NVL
V100 GPU
V100 GPU
V100 GPU
V100 GPU
V100 GPU
V100 GPU
HBM2

NVIDIA.

教育科研行业应用

对于一个包含数百万个颗粒，即使在高性能计算集群上，也可能要花费超过50万 CPU 小时的时间。引入 GPU 加速技术是目前很多主流软件的选择，譬如Relion GPU版，在GPU的加速帮助下，已经大大缩短了分析计算的时间与成本。





近几年来，清华大学研究团队主要与麻省理工学院、加州理工学院和西澳大利亚大学等LSC成员合作开展工作，主要研究成果包括:GPU加速引力波暴数据分析和实现低延迟实时致密双星并合信号的搜寻;采用机器学习方法加强引力波数据噪声的分析;

VASP（VIENNA AB-INITIO SIMULATION PACKAGE）

- VASP是维也纳大学Hafner小组开发的进行电子结构计算和量子力学-分子动力学模拟软件包。它是目前原子尺度材料模拟和计算物质科学研究中最流行的软件之一。

- 采用周期性边界条件处理原子、分子、团簇、纳米线(或管)、薄膜、晶体、准晶和无定性材料的结构参数和构型、状态方程和力学性质、电子结构、光学性质、磁学性质、晶格动力学性质、表面体系的模拟等。

GAUSSIAN

- Gaussian是一个功能强大的量子化学综合软件包。高斯功能：过渡态能量和结构、键和反应能量、分子轨道、原子电荷和电势、振动频率、红外和拉曼光谱、核磁性质、极化率和超极化率、热力学性质、反应路径，计算可以对体系的基态或激发态执行。可以预测周期体系的能量，结构和分子轨道。因此，Gaussian可以用于研究许多化学领域的课题，例如取代基的影响，化学反应机理，势能曲面和激发能等等。

Quantum Chemistry

APPLICATION NAME	COMPANY/DEVELOPER	PRODUCT DESCRIPTION	SUPPORTED FEATURES	GPU SCALING
Abinit	ABINIT	Allows to find total energy, charge density and electronic structure of systems made of electrons and nuclei within DFT.	• Local Hamiltonian • Non-local Hamiltonian • LOBPCG algorithm • Diagonalization/ orthogonalization.	Multi-GPU Single Node
ACES 4	University of Florida	New SIA/aces4 development A new super instruction architecture with interface applications for quantum chemistry (aces4) is now under development. Details and downloads can be found on the project's git hub site.	• Integrating scheduling GPU into SIAL programming language and SIP runtime environment.	Multi-GPU Single Node
ACES III	University of Florida	Takes best features of parallel implementations of quantum chemistry methods for electronic structure.	• Integrating scheduling GPU into SIAL programming language and SIP runtime environment.	Multi-GPU Multi-Node
ADF	Software for Chemistry & Materials	Density Functional Theory (DFT) software package that enables first-principles electronic structure calculations.	• Geometry optimizations and frequency calculations with GGA functionals.	Multi-GPU Single Node
BigDFT	BigDFT	Implements density functional theory by solving the Kohn-Sham equations describing the electrons in a material.	• DFT; Daubechies wavelets, part of Abinit	Multi-GPU Multi-Node
CP2K	CP2K	Program to perform atomistic and molecular simulations of solid state, liquid, molecular and biological systems.	• DBCSR (space matrix multiply library)	Multi-GPU Multi-Node
GAMESS-UK	Open Source	The general purpose ab initio molecular electronic structure program for performing SCF-, DFT- and MCSCF-gradient calculations.	• (ss|ss) type integrals within calculations using Hartree-Fock ab initio methods and density functional theory • Supports organics and inorganics.	Multi-GPU Multi-Node
GAMESS-US	Ames Laboratory/Iowa State University	Computational chemistry suite used to simulate atomic and molecular electronic structure.	• Libqc with Rys Quadrature Algorithm, Hartree-Fock, MP2 and CCSD	Multi-GPU Multi-Node
Gaussian	Gaussian, Inc.	Predicts energies, molecular structures, and vibrational frequencies of molecular systems.	• Joint NVIDIA, PGI and Gaussian collaboration	Multi-GPU Single Node
GPAW	GPAW	Real-space grid DFT code written in C and Python	• Electrostatic poisson equation • Orthonormalizing of vectors • Residual minimization method (rmm-diis)	Multi-GPU Multi-Node
gWL-LSMS	ORNL	Materials code for investigating the effects of temperature on magnetism.	• Generalized Wang-Landau method	Multi-GPU Multi-Node
LATTE	Open Sourcee	Density matrix computations	• CU_BLAS, SP2 Algorithm	Multi-GPU Single Node

VIDIA.

LSDalton	LSDalton	Linear-scaling HF and DFT code suitable for large molecular systems, now also with some CCSD capabilitiesTensor Algebra Library Routines for Shared Memory Systems which is being used to GPU accelerate three (3) CAAR codes: NWChem, LSDALTON and DIRAC.	• (T) correction to the CCSD energy • RI-MP2 energy/gradient (in development) • CCSD energy (in development) • GPU-based ERI generator (in development)	Multi-GPU Single Node
MOLCAS	MOLCAS	Methods for calculating general electronic structures in molecular systems in both ground and excited states.	• CU_BLAS	Multi-GPU Single Node
MOPAC2012	MOPAC	Semiempirical Quantum Chemistry	• Pseudodiagonalization, full diagonalization, and density matrix assembling via Magma libraries	Single GPU Single Node
NWChem	NWChem	NWChem aims to provide its users with computational chemistry tools that are scalable both in their ability to treat large scientific computational chemistry problems efficiently, and in their use of available parallel computing resources from high-performance parallel supercomputers to conventional workstation clusters.Tensor Algebra Library Routines for Shared Memory Systems which is being used to GPU accelerate three (3) CAAR codes: NWChem, LSDALTON and DIRAC.	• Triples part of Reg-CCSD(T), CCSD and EOMCCSD task schedulers	Multi-GPU Single Node
Octopus	Harvard University	Used for ab initio virtual experimentation and quantum chemistry calculations.	• Full GPU support for ground-state, real-time calculations • Kohn-Sham Hamiltonian, orthogonalization, subspace diagonalization, poisson solver, time propagation DFT application	Single GPU Single Node
PEtot	Lawrence Berkeley Laboratories	First principles materials code that computes the behavior of the electron structures of materials.	• Density functional theory (DFT) plane wave pseudopotential calculations	Multi-GPU Single Node
Q-CHEM	Q-Chem Inc.	Computational chemistry package designed for HPC clusters.	• Various features including RI-MP2	Single GPU Single Node
QBox	Open Source	Qbox is a C++/MPI scalable parallel implementation of first-principles molecular dynamics (FPMD) based on the plane-wave, pseudopotential formalism. Qbox is designed for operation on large parallel computers.		Single GPU Single Node
QMCPACK	QMCPACK	QMCPACK, an open-source production level many-body ab initio Quantum Monte Carlo code for computing the electronic structure of atoms, molecules, and solids.	• Main features	Multi-GPU Multi-Node
Quantum Espresso/ PWscf	Quantum Espresso Foundation	An integrated suite of computer codes for electronic structure calculations and materials modeling at the nanoscale.	• PWscf package: linear algebra (matix multiply), explicit computational kernels, 3D FFTs	Multi-GPU Multi-Node
QUICK	Michigan State University	QUICK is a GPU-enabled ab intio quantum chemistry software package.	• Running Hartree-Fock and DFT energy on GPU, Supports s, p, d, f orbitals on energy calculation, HF gradient with s,p,d orbital support, GPU-based ERI generator	Multi-GPU Single Node

RMG	North Carolina State University	RMG is a density functional theory (DFT) based electronics structure code that uses real space grids to represent wavefunctions, charge densities, and ionic potentials. Designed for scalability, it has been run succesfully on systems with thousands of nodes (including GPU nodes) and hundreds of thousands of CPU cores.	• Supports 10k+ GPU nodes, multipetaflops capable • Handles thousands of atoms with full DFT precision • Supports multiple GPUs per node • Fully open source, with installation support, Downloads, documentation, forums www.rmgdft.org • Cray XE6/XK7	Multi-GPU Single Node
TAL-SH	Oak Ridge National Lab	Tensor Algebra Library Routines for Shared Memory Systems which is being used to accelerated three (3) CAAR codes; NWChem, LSDALTON and DIRAC.	• TAL-SH: Tensor Algebra Library for Shared Memory Computers: Nodes equipped with multicore CPU, NVIDIA GPU, and Intel Xeon Phi (in progress).	Multi-GPU Multi-Node
TeraChem	PetaChem LLC	Quantum chemistry software designed to run on NVIDIA GPU.	• Full GPU-based solution; Performance compared to GAMESS CPU version	Multi-GPU Single Node
VASP	University of Vienna	Complex package for performing ab-initio quantum-mechanical molecular dynamics (MD) simulations using pseudopotentials or the projector-augmented wave method and a plane wave basis set.	• Blocked Davidson (ALGO = NORMAL & FAST), RMM-DIIS (ALGO = VERYFAST & FAST), K-Points and optimization for critical step in exact exchange calculations	Multi-GPU Single Node

Tesla vs GeForce

显存ECC校验,多GPU之间RDMA,被动散热, 技术支持

24/7 Uptime

Scalable Performance

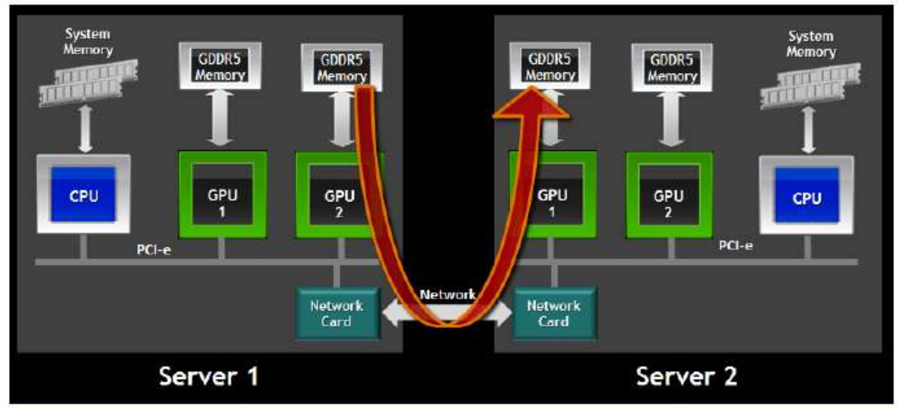
Data Center Ready

GPUDirect RDMA

Direct transfers between GPUs

67% Lower GPU-to-GPU Latency

5x Higher GPU-to-GPU MPI Bandwidth

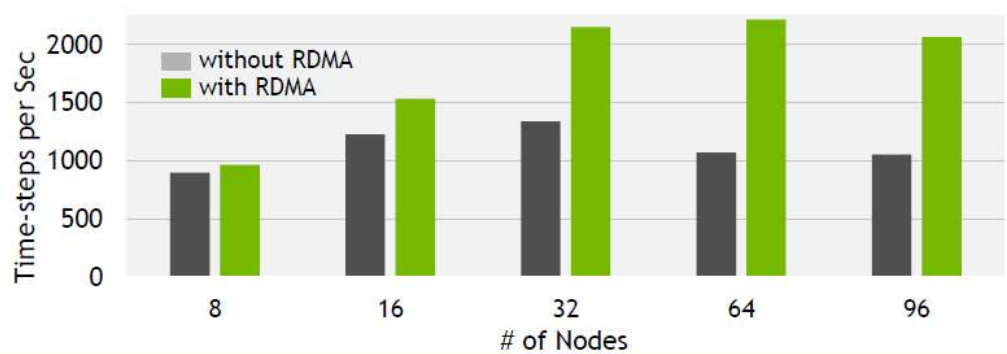


Up to 2x Faster

Application Performance at Scale with GPUDirect RDMA

Hoomd-Blue Application

LJ Liquid Benchmark, 256K Particles




NVIDIA